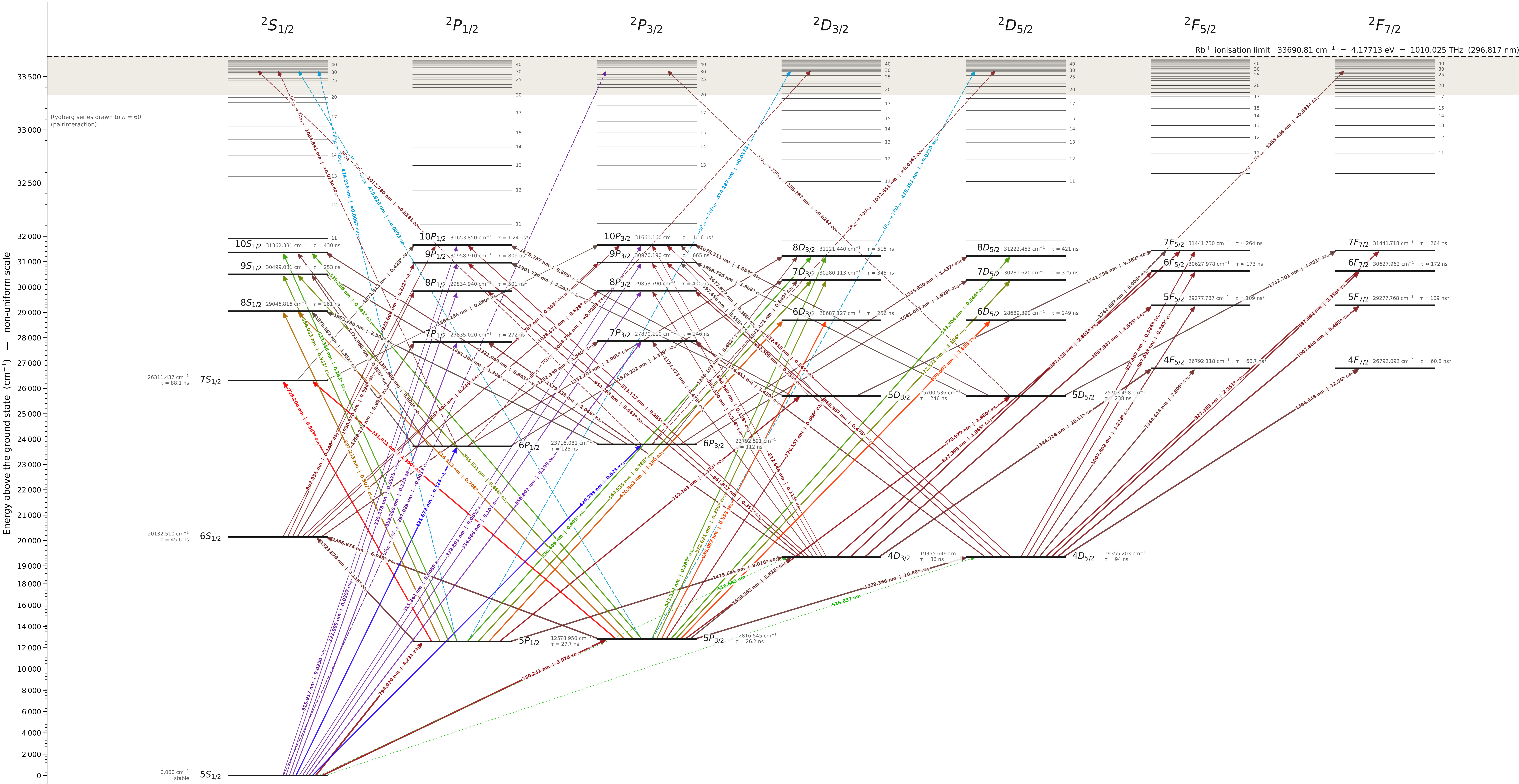
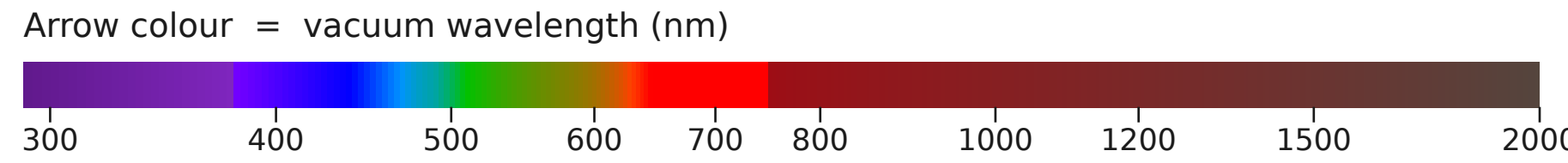


# Rubidium-87

<sup>87</sup>Rb (Rb I), nuclear spin  $I = 3/2$ . Energy levels and transitions below 2  $\mu\text{m}$ : 36 levels, 94 lines.

Arrow label: vacuum wavelength  $\lambda$  | reduced dipole matrix element  $|\langle J||er||J\rangle|$  in  $e a_0$   
Matrix elements are measured or NIST values; \* = high-accuracy theory,  $\approx$  = model calculation.  
Arrow width grows with the Einstein coefficient  $A$ . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: NIST energy and lifetime (measured where a value exists, otherwise ARC, marked  $\approx$ ).



## Key transitions

| Transition                      | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $\tilde{\nu}$ ( $\text{cm}^{-1}$ ) | $ \langle J  er  J\rangle $ ( $e a_0$ ) | $A$ ( $\text{s}^{-1}$ ) | $\Gamma/2\pi$ up. (MHz) | branch (%) | Typical use                                            |
|---------------------------------|--------------------|--------------------|-------------|------------------------------------|-----------------------------------------|-------------------------|-------------------------|------------|--------------------------------------------------------|
| $5S_{1/2} \rightarrow 5P_{3/2}$ | 780.2412           | 780.0266           | 384.2305    | 12816.549                          | 5.978                                   | $3.811\text{e}+07$      | 6.065                   | 100.0      | D2: MOT, imaging, Rydberg lower leg                    |
| $5S_{1/2} \rightarrow 5P_{1/2}$ | 794.9788           | 794.7602           | 377.1075    | 12578.951                          | 4.231                                   | $3.610\text{e}+07$      | 5.746                   | 100.0      | D1: optical pumping, $\Lambda$ / gray-molasses cooling |
| $5S_{1/2} \rightarrow 6P_{3/2}$ | 420.2989           | 420.1805           | 713.2839    | 23792.591                          | 0.523                                   | $1.866\text{e}+06$      | 1.421                   | 21.1       | blue leg of 420 + 1013 Rydberg excitation              |
| $5S_{1/2} \rightarrow 6P_{1/2}$ | 421.6726           | 421.5539           | 710.9602    | 23715.081                          | 0.324                                   | $1.414\text{e}+06$      | 1.273                   | 17.4       | blue leg (421 + 1005)                                  |
| $5P_{3/2} \rightarrow 5D_{5/2}$ | 775.9786           | 775.7651           | 386.3411    | 12886.953                          | 1.980*                                  | $2.833\text{e}+06$      | 0.667                   | 66.8       | 780 + 776 ladder; 5S-5D two-photon at 778.1 nm         |
| $5P_{1/2} \rightarrow 5D_{3/2}$ | 762.1030           | 761.8932           | 393.3752    | 13121.586                          | 1.353*                                  | $2.095\text{e}+06$      | 0.646                   | 51.0       | 795 + 762 ladder                                       |
| $5P_{3/2} \rightarrow 4D_{5/2}$ | 1529.3658          | 1528.9480          | 196.0240    | 6538.658                           | 10.86*                                  | $1.114\text{e}+07$      | 1.693                   | 100.0      | telecom C-band ladder                                  |
| $5P_{3/2} \rightarrow 4D_{3/2}$ | 1529.2615          | 1528.8437          | 196.0374    | 6539.104                           | 3.618*                                  | $1.854\text{e}+06$      | 1.851                   | 15.5       | telecom C-band ladder                                  |
| $5P_{1/2} \rightarrow 4D_{3/2}$ | 1475.6447          | 1475.2415          | 203.1603    | 6776.699                           | 8.016*                                  | $1.013\text{e}+07$      | 1.851                   | 84.5       | telecom S-band ladder                                  |
| $5P_{3/2} \rightarrow 6S_{1/2}$ | 1366.8737          | 1366.5000          | 219.3271    | 7315.965                           | 6.048*                                  | $1.451\text{e}+07$      | 3.493                   | 65.9       | telecom E-band ladder                                  |
| $5P_{1/2} \rightarrow 6S_{1/2}$ | 1323.8791          | 1323.5171          | 226.4500    | 7553.560                           | 4.146*                                  | $7.506\text{e}+06$      | 3.493                   | 34.1       | telecom O-band ladder                                  |
| $5P_{3/2} \rightarrow 6D_{5/2}$ | 630.0068           | 629.8326           | 475.8559    | 15872.845                          | 1.658                                   | $3.712\text{e}+06$      | 0.639                   | 77.9       | red ladder                                             |
| $5P_{3/2} \rightarrow 7S_{1/2}$ | 741.0211           | 740.8171           | 404.5667    | 13494.892                          | 1.350*                                  | $4.536\text{e}+06$      | 1.807                   | 40.0       | ladder to 7S                                           |
| $6S_{1/2} \rightarrow 7P_{3/2}$ | 1292.3904          | 1292.0370          | 231.9674    | 7737.600                           | 1.540*                                  | $5.565\text{e}+05$      | 0.647                   | 14.2       | O-band, from 6S                                        |
| $4D_{5/2} \rightarrow 4F_{7/2}$ | 1344.6483          | 1344.2807          | 222.9523    | 7436.889                           | 12.56*                                  | $1.643\text{e}+07$      | 2.618                   | 98.2       | strongest line of the diagram                          |
| $6P_{3/2} \rightarrow 7D_{5/2}$ | 1541.0626          | 1540.6416          | 194.5362    | 6489.029                           | 1.929*                                  | $3.433\text{e}+05$      | 0.490                   | 11.5       | telecom C-band, from 6P                                |
| $6P_{3/2} \rightarrow 8S_{1/2}$ | 1903.2303          | 1902.7107          | 157.5177    | 5254.225                           | 2.598*                                  | $9.918\text{e}+05$      | 0.989                   | 16.0       | longest 6P line below 2 $\mu\text{m}$                  |

$\lambda$  and  $\nu$  from NIST level energies (centre of gravity, no hyperfine structure);  $\lambda$  air uses standard air (Giddor).  
 $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ . Branch: share of the upper level's decays that takes this line.

## Rydberg excitation wavelengths (vacuum, nm)

| Lower $\rightarrow$ series      | $n = 30$ | $n = 40$ | $n = 50$ | $n = 70$ | $n = 100$ | $n \rightarrow \infty$ | $d_{70}$ pairint. | $d_{70}$ ARC |
|---------------------------------|----------|----------|----------|----------|-----------|------------------------|-------------------|--------------|
| $5S_{1/2} \rightarrow nP_{3/2}$ | 298.113  | 297.510  | 297.247  | 297.029  | 296.918   | 296.817                | 0.0013            | 0.0023       |
| $5P_{1/2} \rightarrow nS_{1/2}$ | 477.099  | 475.483  | 474.788  | 474.216  | 473.927   | 473.667                | 0.0067            | 0.0067       |
| $5P_{1/2} \rightarrow nD_{3/2}$ | 476.682  | 475.318  | 474.707  | 474.187  | 473.917   | 473.667                | 0.0173            | 0.0166       |
| $5P_{3/2} \rightarrow nS_{1/2}$ | 482.570  | 480.916  | 480.205  | 479.620  | 479.324   | 479.059                | 0.0093            | 0.0095       |
| $5P_{3/2} \rightarrow nD_{3/2}$ | 482.143  | 480.747  | 480.122  | 479.591  | 479.315   | 479.059                | 0.0239            | 0.0230       |
| $4D_{5/2} \rightarrow nP_{3/2}$ | 704.765  | 701.404  | 699.946  | 698.736  | 698.121   | 697.564                | 0.0100            | 0.0092       |
| $4D_{5/2} \rightarrow nF_{7/2}$ | 703.548  | 700.913  | 699.701  | 698.649  | 698.092   | 697.564                | 0.0559            | 0.0582       |
| $6S_{1/2} \rightarrow nP_{3/2}$ | 745.611  | 741.850  | 740.219  | 738.866  | 738.178   | 737.556                | 0.0055            | 0.0061       |
| $6P_{1/2} \rightarrow nS_{1/2}$ | 1017.930 | 1010.598 | 1007.464 | 1004.891 | 1003.596  | 1002.433               | 0.0130            | 0.0130       |
| $6P_{1/2} \rightarrow nD_{3/2}$ | 1016.033 | 1009.855 | 1007.100 | 1004.764 | 1003.554  | 1002.433               | 0.0258            | 0.0253       |
| $6P_{3/2} \rightarrow nS_{1/2}$ | 1026.025 | 1018.576 | 1015.393 | 1012.780 | 1011.464  | 1010.283               | 0.0181            | 0.0181       |
| $6P_{3/2} \rightarrow nD_{5/2}$ | 1024.097 | 1017.821 | 1015.023 | 1012.651 | 1011.421  | 1010.283               | 0.0361            | 0.0354       |
| $5D_{3/2} \rightarrow nP_{3/2}$ | 1275.375 | 1264.411 | 1259.681 | 1255.767 | 1253.782  | 1251.986               | 0.0242            | 0.0236       |
| $5D_{5/2} \rightarrow nF_{7/2}$ | 1271.394 | 1262.817 | 1258.887 | 1255.486 | 1253.687  | 1251.986               | 0.0834            | 0.0846       |
| $7S_{1/2} \rightarrow nP_{3/2}$ | 1382.573 | 1369.698 | 1364.148 | 1359.560 | 1357.233  | 1355.129               | 0.0114            | 0.0118       |
| $4F_{7/2} \rightarrow nD_{5/2}$ | 1478.152 | 1465.114 | 1459.322 | 1454.424 | 1451.889  | 1449.545               | 0.0186            | 0.0186       |
| $7P_{3/2} \rightarrow nS_{1/2}$ | 1764.032 | 1742.128 | 1732.838 | 1725.240 | 1721.425  | 1718.006               | 0.0291            | 0.0291       |
| $7P_{3/2} \rightarrow nD_{5/2}$ | 1758.339 | 1739.920 | 1731.758 | 1724.865 | 1721.300  | 1718.006               | 0.0491            | 0.0485       |

Rydberg term energies from pairinteraction, lower level from NIST.  $d_{70} = |\langle J||er||J\rangle|$  to  $n = 70$  in  $e a_0$  (calculated).  
 $S \rightarrow nP$  values come from a near-cancelling integral: treat them as an order of magnitude.

## <sup>87</sup>Rb hyperfine structure ( $I = 3/2$ )

| Level      | A (MHz)      | B (MHz)     | Shift of each F level from the centre of gravity (MHz) |
|------------|--------------|-------------|--------------------------------------------------------|
| $5S_{1/2}$ | 3417.341305  | -           | F=1: -4271.68 F=2: +2563.01                            |
| $5P_{1/2}$ | 406.5(3)     | -           | F=1: -508.10 F=2: +304.86                              |
| $5P_{3/2}$ | 84.720(3)    | 12.4970(20) | F=0: -302.08 F=1: -229.86 F=2: -72.91 F=3: +193.74     |
| $4D_{5/2}$ | -16.786(10)  | 3.82(16)    | F=1: +90.80 F=2: +54.17 F=3: +2.10 F=4: -61.99         |
| $4D_{3/2}$ | 24.75(12)    | 2.19(11)    | F=0: -90.08 F=1: -67.52 F=2: -20.20 F=3: +56.23        |
| $6S_{1/2}$ | 807.35(4)    | -           | F=1: -1009.19 F=2: +605.51                             |
| $6P_{1/2}$ | 132.569(9)   | -           | F=1: -165.71 F=2: +99.43                               |
| $6P_{3/2}$ | 27.706(11)   | 3.97(3)     | F=0: -98.93 F=1: -75.20 F=2: -23.76 F=3: +63.33        |
| $5D_{3/2}$ | 14.4303(5)   | 0.9320(17)  | F=0: -52.95 F=1: -39.45 F=2: -11.52 F=3: +32.70        |
| $5D_{5/2}$ | -7.4605(3)   | 1.2713(20)  | F=1: +40.06 F=2: +24.12 F=3: +1.17 F=4: -27.66         |
| $7S_{1/2}$ | 319.7500(20) | -           | F=1: -399.69 F=2: +239.81                              |
| $7P_{1/2}$ | 59.92(9)     | -           | F=1: -74.90 F=2: +44.94                                |
| $7P_{3/2}$ | 12.570(10)   | 1.762(16)   | F=0: -44.94 F=1: -34.13 F=2: -10.75 F=3: +28.72        |
| $6D_{3/2}$ | 7.84(5)      | 0.53(6)     | F=0: -28.74 F=1: -21.43 F=2: -6.28 F=3: +17.77         |
| $6D_{5/2}$ | -3.61(6)     | -0.20(20)   | F=1: +18.81 F=2: +11.75 F=3: +1.01 F=4: -13.59         |
| $7D_{3/2}$ | 4.53(3)      | 0.26(4)     | F=0: -16.66 F=1: -12.39 F=2: -3.59 F=3: +10.26         |
| $7D_{5/2}$ | -1.98(28)    | -           | F=1: +10.39 F=2: +6.43 F=3: +0.49 F=4: -7.42           |

A, B: magnetic-dipole and electric-quadrupole constants (measured; see data file for sources).